python project with Jenkins

- First to create aws instance Install java

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-25-57:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
```

i-0b68928f3f6d95c4b (py-jenkins)

PublicIPs: 65.1.242.169 PrivateIPs: 172.31.25.57

- A er install java, install Jenkins

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-122.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-122.04, mixed mode, sharing)
ubuntu@ip-172-31-25-57:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
--2026-02-17 15:56:43--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc     100%[=================================================================
```

- A er install Jenkins, install docker.io

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-25-57:~$ sodu apt install docker.io
sodu: command not found
ubuntu@ip-172-31-25-57:~$ sudo apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 46 not upgraded.
Need to get 76.3 MB of archives.
After this operation, 289 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

- A er installing all the needed package
- Start the packages
- Start Jenkins

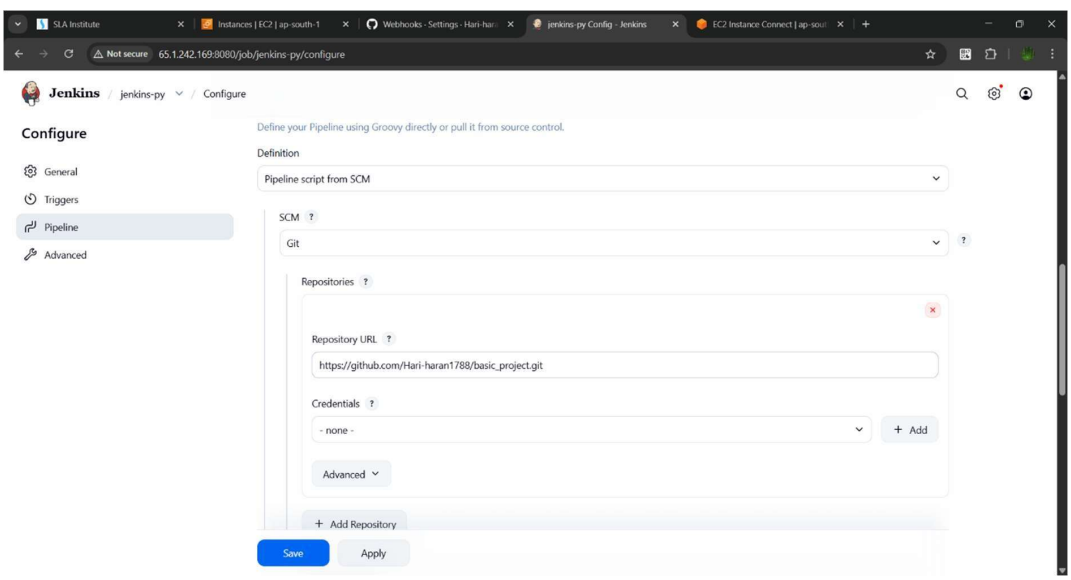
```
ubuntu@ip-172-31-25-57:~$ sudo su
root@ip-172-31-25-57:/home/ubuntu# systemctl start jenkins
root@ip-172-31-25-57:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-02-17 15:57:31 UTC; 2min 16s ago
   Main PID: 5594 (java)
      Tasks: 43 (limit: 4556)
     Memory: 593.9M
        CPU: 13.353s
     CGroup: /system.slice/jenkins.service
             └─5594 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Feb 17 15:57:26 ip-172-31-25-57 jenkins[5594]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Feb 17 15:57:26 ip-172-31-25-57 jenkins[5594]: [LF]>
Feb 17 15:57:26 ip-172-31-25-57 jenkins[5594]: [LF]> ************************************************************
Feb 17 15:57:26 ip-172-31-25-57 jenkins[5594]: [LF]> ************************************************************
```

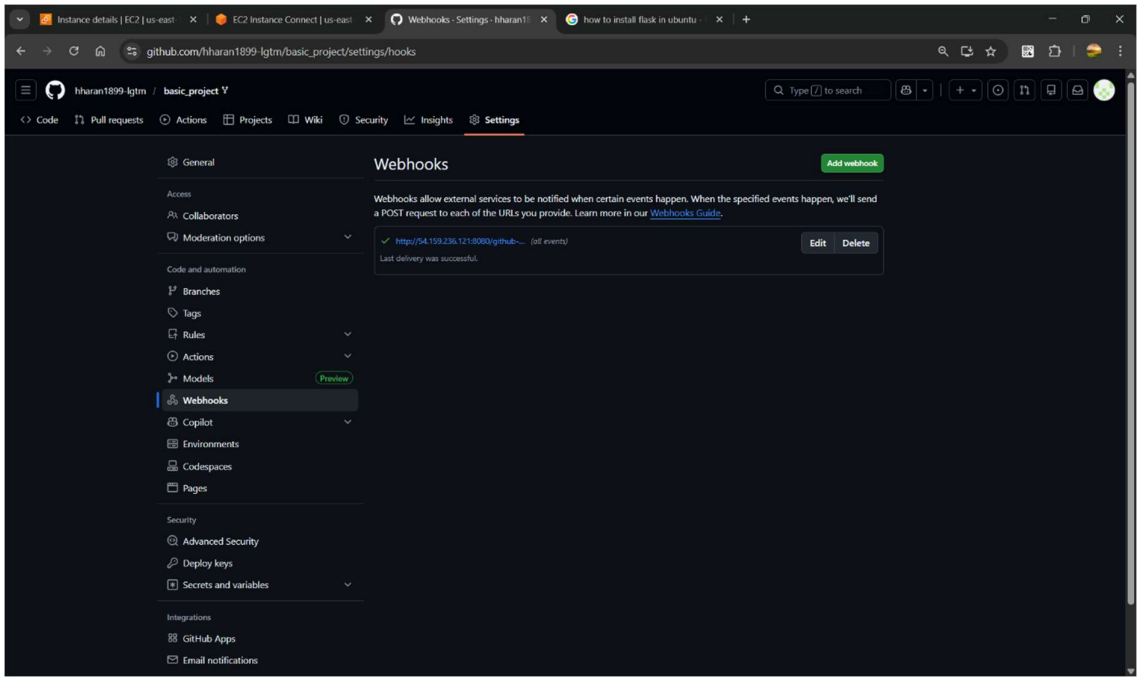
- Start docker

```
ubuntu@ip-172-31-25-57:~$ sudo su
root@ip-172-31-25-57:/home/ubuntu# systemctl start docker
root@ip-172-31-25-57:/home/ubuntu# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2026-02-17 15:58:12 UTC; 2min 51s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 6275 (dockerd)
      Tasks: 9
     Memory: 20.6M
        CPU: 227ms
     CGroup: /system.slice/docker.service
             └─6275 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

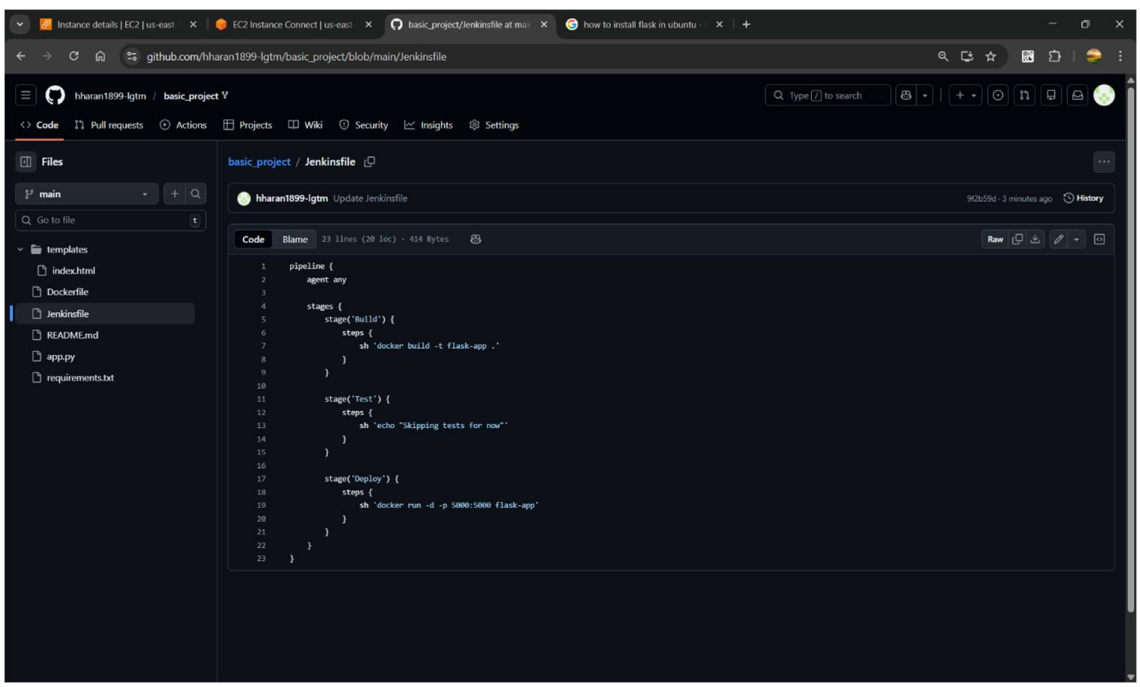
- The Jenkins is running and to create the pipeline named ”Jenkins-py”.
- While crea ng pipeline add the “git repo url” and change the branch name as “main”



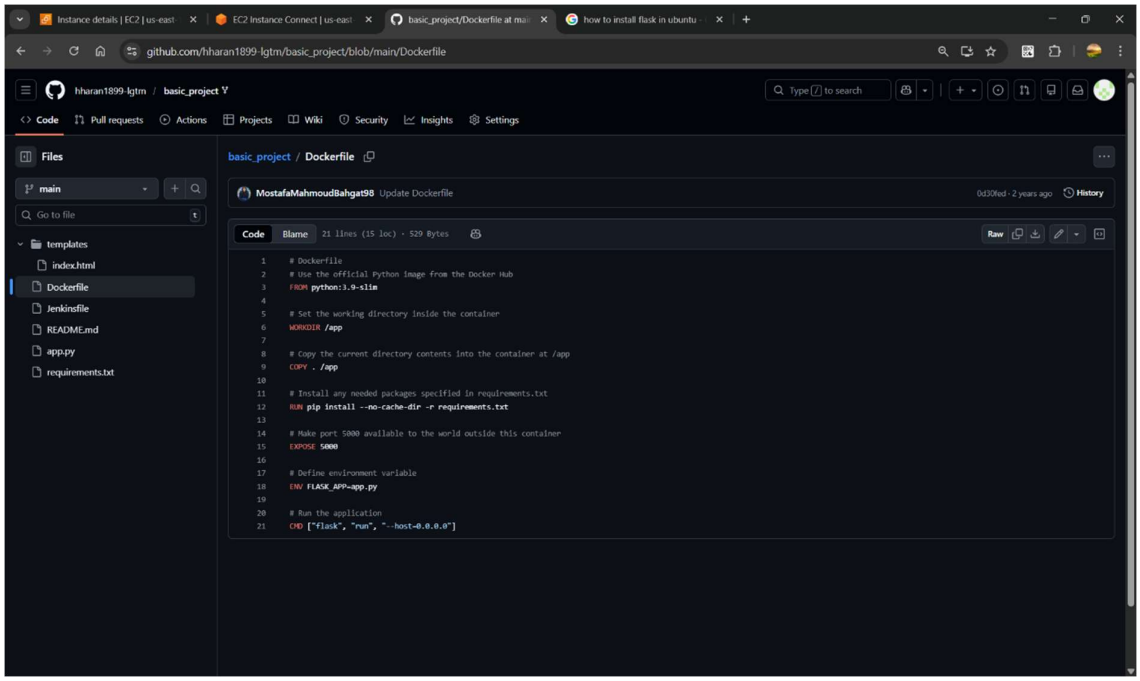
- Create the webhook in the repository



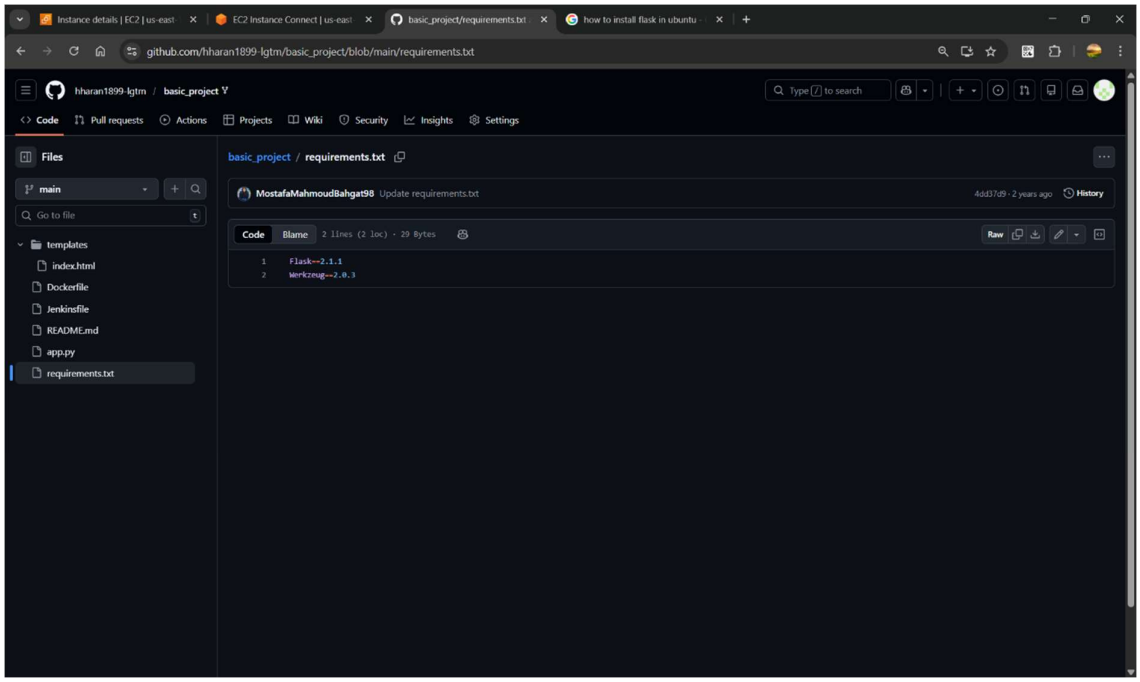
- The repofiles are;
- Jenkinsfile



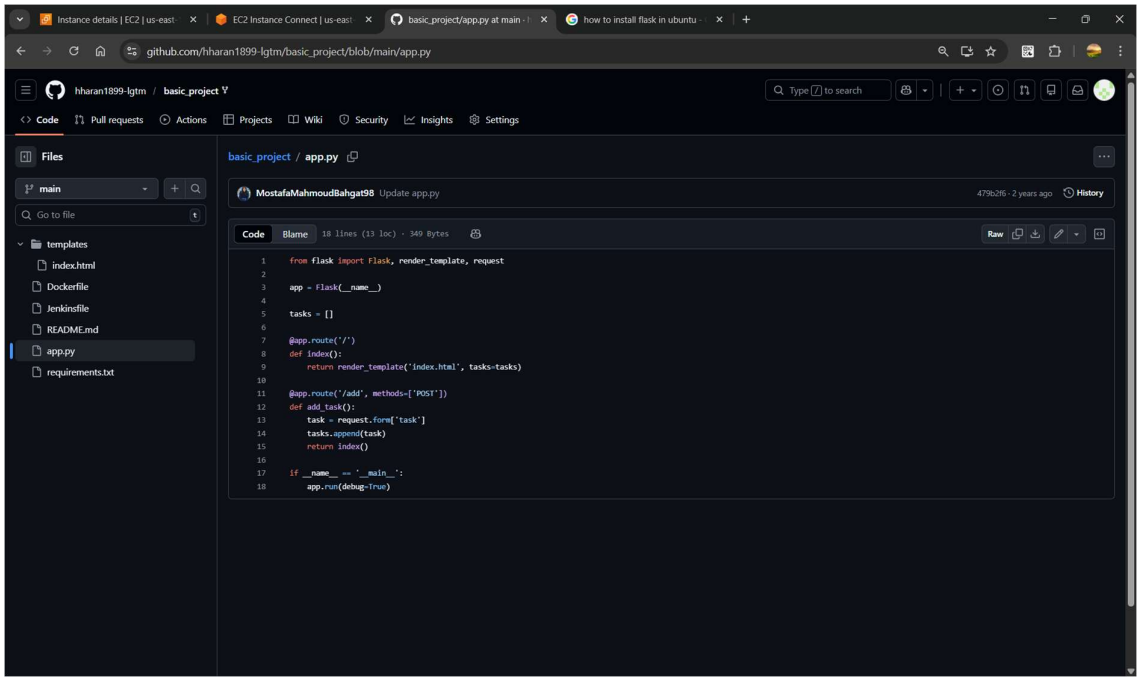
- Dockerfile



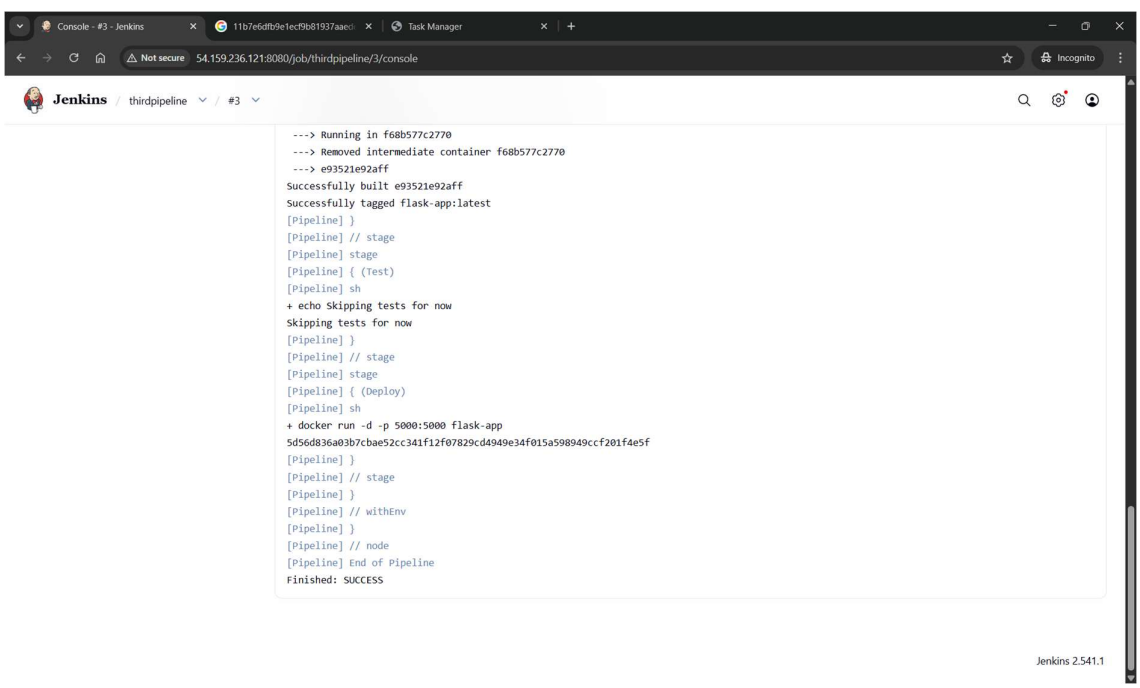
- requirement.txt

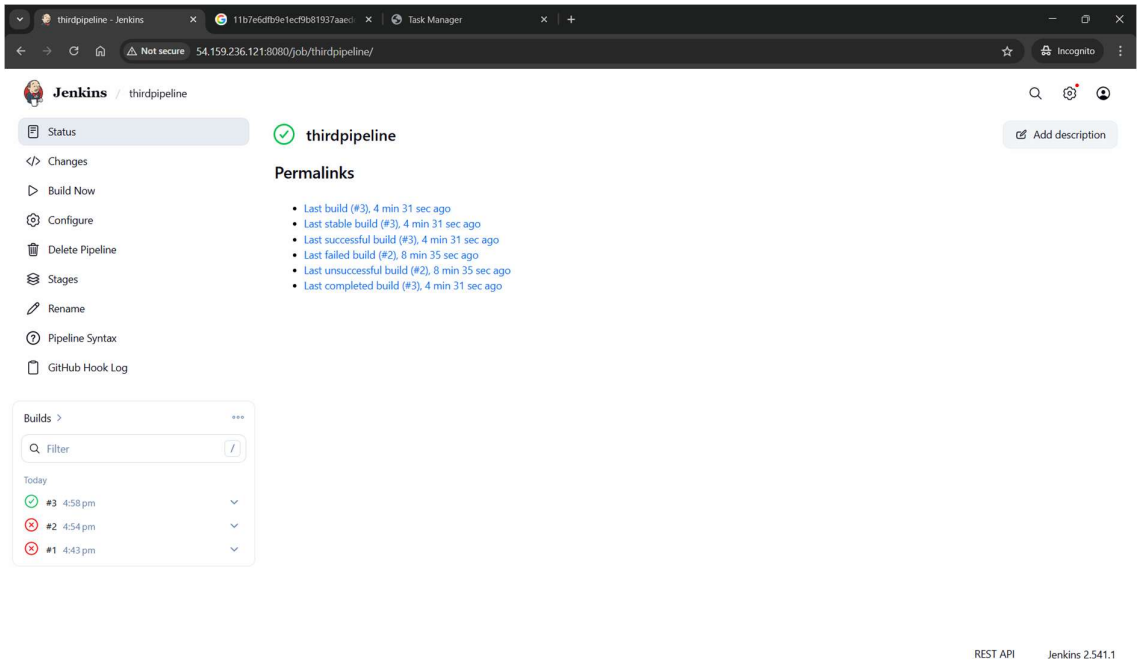
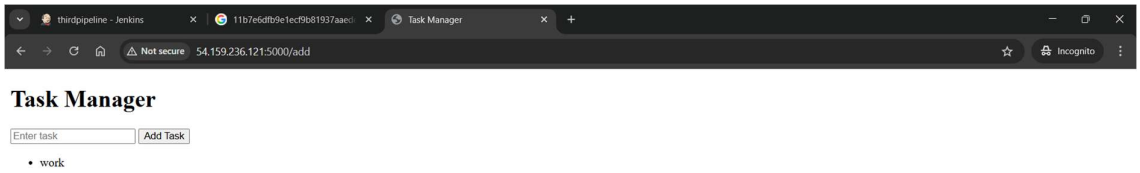


- app.py



- A er connec ng the file the pipeline is automa cally triggered □ The pipeline in success





- The output is hosted by docker container